CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 18: Tangents and Intersecting Chords

Topic-wise short notes with solved examples

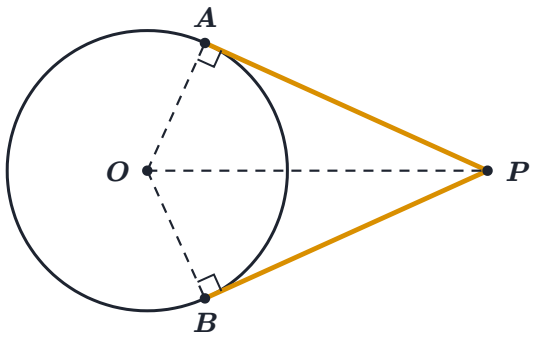
A tangent touches a circle at exactly one point. This chapter covers the properties of tangents, circles that touch each other, the products formed by intersecting chords, and the alternate segment theorem.

1 Tangent and secant

- A **secant** is a line that cuts a circle at two points. A **tangent** is a line that touches the circle at exactly one point, the **point of contact**.
- From a point **inside** a circle no tangent can be drawn; from a point **on** the circle, exactly one; from a point **outside**, exactly two.

2 Tangent properties

- **Theorem 1**: the tangent at any point of a circle and the radius through that point are **perpendicular** to each other.
- **Theorem 2**: if two tangents are drawn to a circle from an external point, then (i) they are **equal in length**, (ii) they subtend equal angles at the centre, and (iii) they are equally inclined to the line joining the point to the centre.
- For tangents PA and PB from P to a circle with centre O: $\angle APB + \angle AOB = 180^\circ$.
- Length of the tangent from a point at distance d from the centre: $\sqrt{d^2 - r^2}$.



Tangents PA and PB from an external point P, each perpendicular to the radius at the point of contact

Example 2.1 A point P is 17 cm from the centre of a circle of radius 8 cm. Find the length of the tangent from P.

SOLUTION

The radius is perpendicular to the tangent at the point of contact T, so $PT^2 = OP^2 - OT^2$.

$PT = \sqrt{17^2 - 8^2} = \sqrt{225}$

$PT = 15$ **cm**

Example 2.2 PA and PB are tangents to a circle with centre O, and $\angle APB = 70^\circ$. Find $\angle AOB$ and $\angle OAB$.

SOLUTION

$\angle AOB = 180^\circ - 70^\circ = 110^\circ$

$OA = OB$, so $\angle OAB = \dfrac{180^\circ - 110^\circ}{2}$

$\angle AOB = 110^\circ$ **and** $\angle OAB = 35^\circ$

Example 2.3 A circle is inscribed in $\triangle ABC$, touching AB, BC and CA at P, Q and R. If $AP = 4$ cm, $BP = 6$ cm and $CR = 5$ cm, find the perimeter of the triangle.

SOLUTION

Tangents from a point are equal: $AR = AP = 4$, $BQ = BP = 6$, $CQ = CR = 5$.

Perimeter $= 2(4 + 6 + 5)$

Perimeter $= 30$ **cm**

3 Circles that touch each other

- If two circles touch, the **point of contact lies on the line joining their centres**.

The circles touch	Distance between the centres	Common tangents
Externally	$R + r$	3
Internally	$R - r$	1

- Two circles that do not meet and lie outside each other have 4 common tangents; two intersecting circles have 2.

Example 3.1 Two circles of radii 5 cm and 3 cm touch each other. Find the distance between their centres if they touch (i) externally (ii) internally.

SOLUTION

(i) $5 + 3$

(ii) $5 - 3$

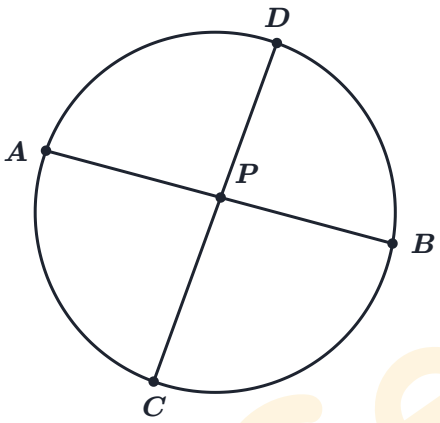
(i) 8 cm (ii) 2 cm

4 Intersecting chords

- **Theorem 3**: if two chords of a circle intersect, inside or outside the circle when produced, the product of the lengths of the segments of one chord equals the product of the lengths of the segments of the other.

$$PA \times PB = PC \times PD$$

- Here P is the point of intersection of the chords AB and CD, and every length is measured **from** P.



Chords AB and CD meet at P: PA × PB = PC × PD

Example 4.1 Chords AB and CD of a circle intersect at a point P inside it. If $PA = 4$ cm, $PB = 6$ cm and $PC = 3$ cm, find PD.

SOLUTION

$PA \times PB = PC \times PD$

$4 \times 6 = 3 \times PD$

$PD = 8$ **cm**

Example 4.2 Chords AB and CD, when produced, meet at a point P outside the circle. If $PA = 12$ cm, $AB = 7$ cm and $PC = 10$ cm, find CD. (B lies between P and A, and D between P and C.)

SOLUTION

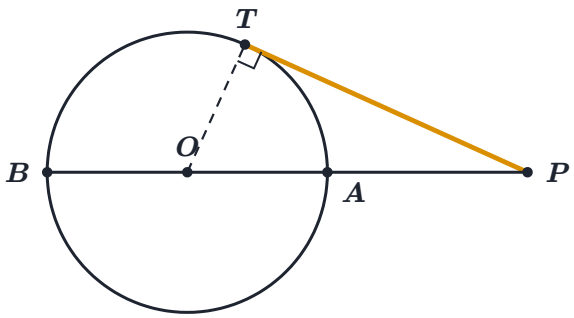
$PB = PA - AB = 12 - 7 = 5$ cm

$PA \times PB = PC \times PD$: $12 \times 5 = 10 \times PD$, so $PD = 6$ cm

$CD = PC - PD = 4$ **cm**

5 Tangent and secant from an external point

- **Theorem 4**: if a chord AB and a tangent at T meet at an external point P, then



Tangent PT and secant PAB from the external point P

$$PT^2 = PA \times PB$$

Example 5.1 From an external point P, a tangent PT and a secant PAB are drawn to a circle. If $PA = 4$ cm and $AB = 5$ cm, find PT.

SOLUTION

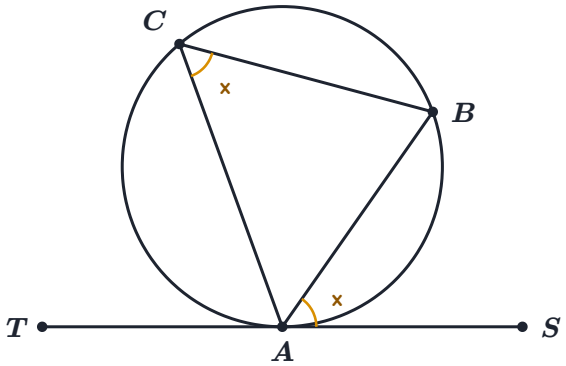
$PB = PA + AB = 9$ cm

$PT^2 = PA \times PB = 4 \times 9 = 36$

$PT = 6$ cm

6 The alternate segment theorem

- **Theorem 5**: the angle between a tangent and a chord through the point of contact is equal to the angle in the **alternate segment**.
- If TAS is the tangent at A and AB is a chord, then $\angle BAS$ equals the angle that AB subtends at any point of the circle on the side of AB away from S.



Angle BAS between the tangent and the chord equals angle ACB in the alternate segment

Example 6.1 TAS is a tangent to a circle at A, and AB is a chord with $\angle BAS = 55°$. C is a point on the circle in the alternate segment. Find $\angle ACB$.

SOLUTION

The angle between the tangent and the chord equals the angle in the alternate segment.

$\angle ACB = 55°$

Example 6.2 PQ is a tangent to a circle at A. AB is a chord and C is a point on the circle such that $\angle ACB = 48°$ and $\angle ABC = 70°$. Find $\angle BAQ$ and $\angle CAP$, where B is on the side of Q.

SOLUTION

$\angle BAQ = \angle ACB = 48°$ (alternate segment)

$\angle CAP = \angle ABC = 70°$ (alternate segment)

$\angle BAQ = 48°$ **and** $\angle CAP = 70°$**. (Check:** $48° + 62° + 70° = 180°$**, since** $\angle BAC = 62°$**.)**

7 Common mistakes

- Measuring a segment from the wrong end. In $PA \times PB$ both lengths start at P.
- For external intersection, using AB instead of PB in the product.
- Writing $PT = PA \times PB$. It is PT^2.
- Choosing the angle in the same segment instead of the alternate one.

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